Yuyuan Han

Computational imaging & robust image-free sensing

hanyuyuan6@gmail.com · yuyuan.han@mail.polimi.it · Milan, Italy
hanyuyuan6.github.io · GitHub / Hanyuyuan6 · Google Scholar · ORCID / 0009-0005-7643-6331

PhD student in computational imaging and robust image-free sensing: single-pixel measurement design and reconstruction (MATLAB) and direct segmentation from measurements (PyTorch).

Education & Research Appointments

Xi'an Jiaotong University · PhD student — 2022–present
Electronic Science and Technology · Master's-to-PhD track · Advisor: Huaibin Zheng

Politecnico di Milano · Visiting PhD student — Oct 2026–present
Department of Physics · Supervisor: Antonio Pifferi · Planned research: time-domain diffuse optics

Northwest University · B.Eng. — 2018–2022
Electronic Information Science and Technology

Selected Research

The Lift Spectrum · Robust image-free sensing — Preprint

arXiv · 2026 · Paper · Code

- Framed the measurement-to-space lift as the key design question in image-free single-pixel segmentation; designed the STSF network and the TPLS training schedule.
- In simulation at 3.13% sampling: +3.2–9.9 pp foreground mIoU over SPIFS; at 20 dB SNR, +3.1–21.7 pp over the best clean-trained reconstruct-then-segment baseline, which wins without noise.

Sparse Hadamard single-pixel imaging

Optics Express · 2025 · Paper · Code

- Designed tunable, orthogonal sparse Hadamard patterns with two training-free modes and a noise model; more resilient to noise than conventional patterns in 20 and 30 dB SNR simulations, validated optically.

LDGI · Ghost imaging for optical phased-array line patterns

Optics Communications · 2025 · Paper · Code

- Developed loop differential ghost imaging, which corrects the noise asymmetry of line patterns from 1D optical phased arrays; orders-of-magnitude gains over traditional ghost imaging, validated with an SLM.

Current Research

Acquisition-aware single-pixel sensing on an optical bench (manuscript in preparation); question answering from measurements with a frozen vision–language model (exploratory).

Open Source & Tools

Open source · remote-gpu-trainer, an agent skill for Claude Code and Codex that runs, verifies and delivers GPU experiments (60+ GitHub stars)
Tools & lab · PyTorch, MATLAB, COMSOL; DMD/SLM illumination, single-pixel detection, optical alignment and system integration
Device modeling · Geometry optimization of electro-optic phased arrays for beam steering

Award & Service

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling · Code
Reviewer · IEEE TIP, IEEE TCI, Optics Express, Optics Letters (13 verified reviews, 2025–2026)